Appendix

Statement of AI Usage:

ChatGPT (a Large Language Model) was used in a supportive capacity at author's discretion for grammar check, formatting, debugging, retrieving real peer-reviewed articles (through search and deep research function, with manual verification for relevance), and obtaining documents/ guidelines related to coding and modelling techniques in SAS and R.

Appendix A: Tables and Figures:

Table A.1 Odds Ratio of Model 1 without Employment Status and Socioeconomic status

Variable	Category Comparison	OR	95% CI
Alcohol Consumption Frequency (ALCDAYSYR_cate)	**Daily drinker vs Non-drinker**	0.837	(0.760, 0.922)
	Frequent drinker vs Non-drinker	0.788	(0.723, 0.860)
	Moderate drinker vs Non-drinker	0.604	(0.552, 0.661)
	Sometime Drinker vs Non-drinker	0.694	(0.656, 0.733)
	Unknown vs Non-drinker	0.560	(0.526, 0.597)
Age Group (Age_cate)	**30-39 vs 18-29**	0.810	(0.764, 0.860)
	40-49 vs 18-29	0.893	(0.838, 0.952)
	50-59 vs 18-29	0.915	(0.862, 0.973)
	60-69 vs 18-29	0.763	(0.717, 0.811)
	70-79 vs 18-29	0.621	(0.575, 0.670)
	80-85 vs 18-29	0.613	(0.560, 0.672)
	Missing vs 18-29	0.668	(0.268, 1.662)
Sex (Sex_cate)	**Female vs Male**	1.574	(1.516, 1.634)
	Missing vs Male	2.582	(0.252, 26.436)
Race (Race_cate)	**American Indian/Alaska Native vs White only**	0.977	(0.808, 1.181)
	Asian only vs White only	0.535	(0.485, 0.590)
	Black/African American only vs White only	0.765	(0.722, 0.811)

	Missing vs White only	0.658	(0.559, 0.775)
	Multiple Race (1999-2018: Incl vs White only	1.324	(1.150, 1.525)
	Other Race vs White Only	1.150	(0.826, 1.601)
Health Insurance Status (HINOTCOVE_cate)	**Missing vs Yes, has no coverage**	0.834	(0.604, 1.150)
	No, has coverage vs Yes, has no coverage	0.790	(0.748, 0.833)
Hours of Sleep (HRSLEEP_cate)	**Long vs Short**	0.655	(0.612, 0.702)
	Normal vs Short	0.353	(0.334, 0.374)

Table A.2 Odds Ratio of Model 3 with Employment Status and Socioeconomic status but treating missingness as separated category:

Variable	Category Comparison	OR	95% CI
Alcohol Consumption Frequency (ALCDAYSYR_cate)	**Daily drinker vs Non-drinker**	1.043	(0.937, 1.067)
	Frequent drinker vs Non-drinker	1.064	(0.923, 1.046)
	Moderate drinker vs Non-drinker	0.801	(0.640, 0.729)
	Sometime Drinker vs Non-drinker	0.845	(0.426, 0.502)
	Unknown vs Non-drinker	0.572	(0.383, 0.465)
Age Group (Age_cate)	**30-39 vs 18-29**	0.885	(0.274, 1.679)
	40-49 vs 18-29	1.000	(1.443, 1.557)
	50-59 vs 18-29	0.983	(0.224, 23.374)
	60-69 vs 18-29	0.683	(0.736, 1.087)
	70-79 vs 18-29	0.462	(0.493, 0.602)
	80-85 vs 18-29	0.422	(0.648, 0.727)
	Missing vs 18-29	0.679	(0.561, 0.780)
Sex	**Female vs Male**	1.499	(1.076, 1.429)

(Sex_cate)			
	Missing vs Male	2.287	(0.805, 1.577)
Race (Race_cate)	**American Indian/Alaska Native vs White only**	0.894	(0.528, 0.582)
	Asian only vs White only	0.545	(1.016, 1.319)
	Black/African American only vs White only	0.687	(0.742, 1.432)
	Missing vs White only	0.661	(0.908, 1.017)
	Multiple Race (1999-2018: Incl vs White only	1.240	(1.780, 2.077)
	Other Race vs White Only	1.126	(0.684, 0.815)
Employment Status(EMPSTATIMP5_cate)	**Employed vs Unemployed**	0.554	(1.158, 1.351)
	Missing vs Unemployed	1.158	(0.787, 0.989)
Health Insurance Status (HINOTCOVE_cate)	**Missing vs Yes, has no coverage**	1.031	(0.605, 0.708)
	No, has coverage vs Yes, has no coverage	0.961	(0.369, 0.414)
SES(Total Family Income) INCFAM97ON2_cate	**$0 - $34,999 vs $75,000 - $99,999**	1.923	(0.937, 1.067)
	$100,000 and over vs $75,000 - $99,999	0.747	(0.923, 1.046)
	$35,000 - $74,999 vs $75,000 - $99,999	1.251	(0.640, 0.729)
	Missing vs $75,000 - $99,999	0.882	(0.426, 0.502)
Hours of Sleep (HRSLEEP_cate)	**Long vs Short**	0.654	(0.383, 0.465)
	Normal vs Short	0.391	(0.274, 1.679)

Figure A.1 Odds Ratio Association Between Alcohol Consumption Frequency and Depression (without Employment Status and SES variable)

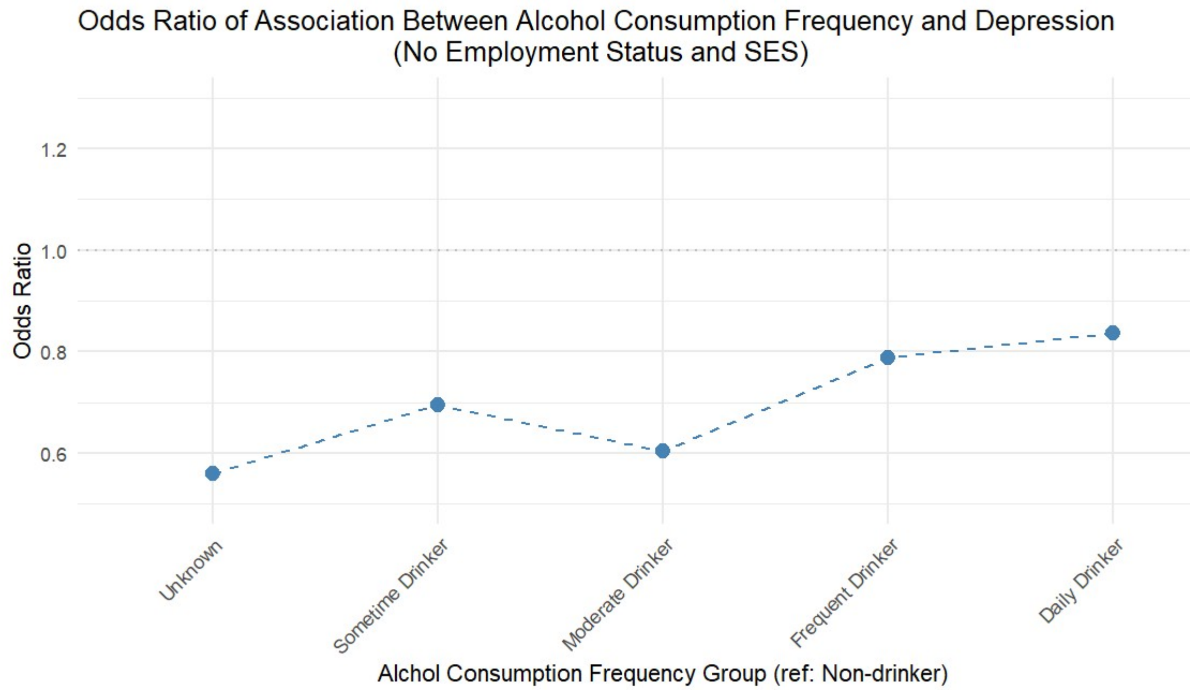
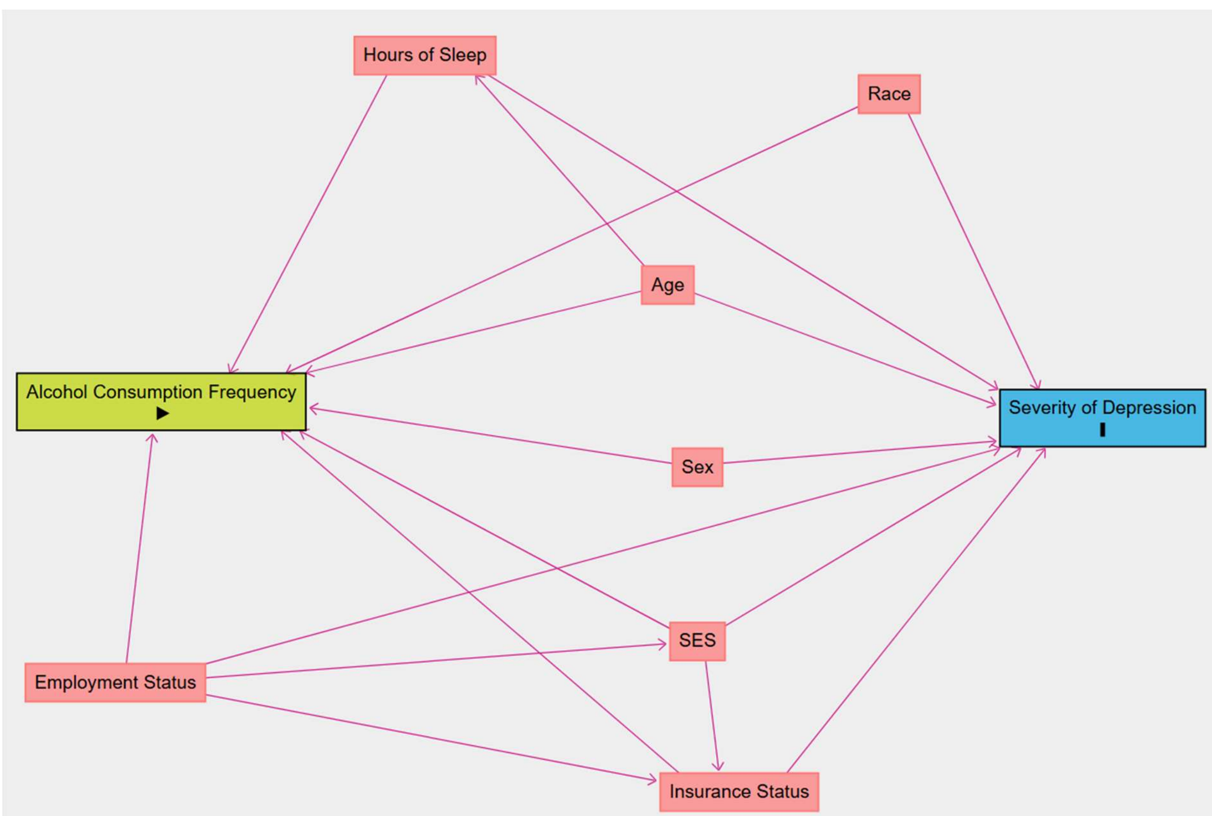


Figure A.2 DAG Plot of Association Exploration between Variables



Appendix B. Codes and outputs

DAG Plot Codes:

```
dag {
"Alcohol Consumption Frequency" [exposure,pos="-1.166,-0.233"]
"Employment Status" [pos="-1.040,0.612"]
"Hours of Sleep" [pos="-0.598,-1.640"]
"Insurance Status" [pos="-0.044,1.242"]
"Severity of Depression" [outcome,pos="0.862,-0.165"]
Age [pos="-0.075,-0.681"]
Race [pos="0.402,-1.479"]
SES [pos="-0.011,0.808"]
Sex [pos="-0.011,0.080"]
"Employment Status" -> "Alcohol Consumption Frequency"
"Employment Status" -> "Insurance Status"
"Employment Status" -> "Severity of Depression"
"Employment Status" -> SES
"Hours of Sleep" -> "Alcohol Consumption Frequency"
"Hours of Sleep" -> "Severity of Depression"
"Insurance Status" -> "Alcohol Consumption Frequency"
"Insurance Status" -> "Severity of Depression"
Age -> "Alcohol Consumption Frequency"
Age -> "Hours of Sleep"
Age -> "Severity of Depression"
Race -> "Alcohol Consumption Frequency"
Race -> "Severity of Depression"
SES -> "Alcohol Consumption Frequency"
SES -> "Insurance Status"
SES -> "Severity of Depression"
Sex -> "Alcohol Consumption Frequency"
Sex -> "Severity of Depression"
}
```

Code of Association Trend (R):

OR of Model with Employment and SES Variables, only fit on non-missing rows

```
> # Load  package
> library(ggplot2)
>
> # Create the data
> OR <- data.frame(
+   Group = c("Unknown", "Sometime Drinker", "Moderate Drinker", "Frequent Dr
inker", "Daily Drinker"),
+   OR = c(0.568, 0.849, 0.796, 1.066, 1.058)
+ )
>
> #factor levels
> OR$Group <- factor(OR$Group, levels = OR$Group)
>
> # Generating Plot
> ggplot(OR, aes(x = Group, y = OR)) +
+   geom_point(size = 3, color = "steelblue") +
+   geom_line(group = 1, color = "steelblue", linetype = "dashed") +
+   geom_hline(yintercept = 1, linetype = "dotted", color = "gray") +
+   labs(
+     title = "Odds Ratio of Association Between Alcohol Consumption Frequenc
y and Depression
+                         (With Employment Status and SES on Non-Missing Rows)
",
+     x = "Alchol Consumption Frequency Group (ref: Non-drinker)",
+     y = "Odds Ratio"
```

```
+   ) +
+   scale_y_continuous(limits = c(0.5, 1.3)) +
+   theme_minimal() +
+   theme(axis.text.x = element_text(angle = 45, hjust = 1))
>
>
```

OR of Model without Employment and SES on Full Data Set

```
> #not that obvious trend version, plus all dip below 1
>
>
> # Create the data set
> OR_model1 <- data.frame(
+   Group = c("Unknown", "Sometime Drinker", "Moderate Drinker", "Frequent Dr
inker", "Daily Drinker"),
+   OR = c(0.560, 0.694, 0.604, 0.788, 0.837)
+ )
>
> #factor levels
> OR_model1$Group <- factor(OR_model1$Group, levels = OR_model1$Group)
>
> # Generating Plot
> ggplot(OR_model1, aes(x = Group, y = OR)) +
+   geom_point(size = 3, color = "steelblue") +
+   geom_line(group = 1, color = "steelblue", linetype = "dashed") +
+   geom_hline(yintercept = 1, linetype = "dotted", color = "gray") +
+   labs(
+     title = "Odds Ratio of Association Between Alcohol Consumption Frequenc
y and Depression
+                                                (No Employment Status and SES)",
+     x = "Alchol Consumption Frequency Group (ref: Non-drinker)",
+     y = "Odds Ratio"
+   ) +
+   scale_y_continuous(limits = c(0.5, 1.3)) +
+   theme_minimal() +
+   theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

SAS Code of Analysis and Outputs:

```
libname Cate "C:\SASLib\thesis";

%macro QuickFreq(var);
proc freq data=NHIS;
table &var;
run;
%mend QuickFreq;

%macro SurveyFreq(var,set);
proc surveyfreq data=&set;
strata Strata;
cluster psu;
weight SAMPWEIGHT;
table &var;
run;
```

```
%mend;

%macro OutTablePerc(set);
data &set;
set &set;
freq_perc=catx('
',put(Frequency,10.),'('||strip(put(RowPercent,8.2))||'%)');
cate_perc=catx('
',put(Frequency,10.),'('||strip(put(Percent,8.2))||'%)');
run;
%mend;


%macro OnlyNeed(set);
data &set;
set &set;
keep keep ALCDAYSYR_cate SLEEPREST_cate freq_perc cate_perc;
run;

%Mend;



data NHIS;
set Cate.nhis_00006;
run;

/*Cleaning and collapsing categories without removing missing*/
data NHIS ;
set NHIS;

where age>=18 and year>=2010;

if Age=997 or Age=999 then Age=.;

/*Recode severity of depression*/
if DEPFEELEVL=0 or DEPFEELEVL=7 or
DEPFEELEVL=8 or DEPFEELEVL=9 then DEPFEELEVL=.;

/*Recode employment status*/
if EMPSTATIMP5=0 then EMPSTATIMP5=.;


/*Recode insurance status*/

if HINOTCOVE=0 or HINOTCOVE=7 or
HINOTCOVE=8 or HINOTCOVE=9 then HINOTCOVE=.;

/*Recode sleep quality*/
if SLEEPREST=96 or SLEEPREST=97
or SLEEPREST=98 or SLEEPREST=99 then SLEEPREST=.;
```

```
/*Recode drinking frequency*/
if ALCDAYSYR=995 or ALCDAYSYR=996
or ALCDAYSYR=997 or ALCDAYSYR=998
or ALCDAYSYR=999 then ALCDAYSYR=.;

if SMOKFREQNOW=0 or SMOKFREQNOW=7
or SMOKFREQNOW=8 or SMOKFREQNOW=9
then SMOKFREQNOW=.;

if RACENEW=997 or RACENEW=998 or RACENEW=999
or RACENEW=530 then RACENEW=.;
if RACENEW=500 or RACENEW=510 then RACENEW=520;

if INCFAM97ON2=97 or INCFAM97ON2=98 or INCFAM97ON2=99
then INCFAM97ON2=.;

if SEX=7 or SEX=8 OR SEX=9
THEN SEX=.;
run;


/*See if alcohol consumption frequency is affected by health
insurance coverage status*/

data NHIS;
   set NHIS;

   length ALCDAYSYR_cate EMPSTATIMP5_cate AGE_cate DEPFEELEVL_cate
          HINOTCOVE_cate INCFAM97ON2_cate RACENEW_cate SEX_cate
          SLEEPREST_cate SMOKFREQNOW_cate $30.;

   if missing(ALCDAYSYR_cate) then ALCDAYSYR_cate = "Missing";

   if missing(EMPSTATIMP5) then EMPSTATIMP5_cate = "Missing";
   else if EMPSTATIMP5 = 1 then EMPSTATIMP5_cate = "Employed";
   else if EMPSTATIMP5 = 2 then EMPSTATIMP5_cate = "Unemployed";
   else EMPSTATIMP5_cate = "Other";

   if missing(AGE) then AGE_cate = "Missing";
   else AGE_cate = put(AGE, best.);

   if missing(DEPFEELEVL) then DEPFEELEVL_cate = "Missing";
   else DEPFEELEVL_cate = put(DEPFEELEVL, best.);

   if missing(HINOTCOVE) then HINOTCOVE_cate = "Missing";
   else HINOTCOVE_cate = put(HINOTCOVE, best.);

   if missing(INCFAM97ON2) then INCFAM97ON2_cate = "Missing";
   else INCFAM97ON2_cate = put(INCFAM97ON2, best.);

   if missing(RACENEW) then RACENEW_cate = "Missing";
   else RACENEW_cate = put(RACENEW, best.);
```

```
   if missing(SEX) then SEX_cate = "Missing";
   else SEX_cate = put(SEX, best.);

   if missing(SLEEPREST) then SLEEPREST_cate = "Missing";
   else SLEEPREST_cate = put(SLEEPREST, best.);

if missing(SMOKFREQNOW) then SMOKFREQNOW_cate = "Missing";
else if SMOKFREQNOW = 1 then SMOKFREQNOW_cate = "None-Smoker";
else if SMOKFREQNOW = 2 then SMOKFREQNOW_cate = "Some Day Smoker";
else if SMOKFREQNOW = 3 then SMOKFREQNOW_cate = "Everyday Smoker";
else if SMOKFREQNOW = 0 and smokev = 1 then SMOKFREQNOW_cate = "None-
Smoker"; /* NIU handling */
else SMOKFREQNOW_cate = "Unknown";

run;



data NHIS;
set NHIS;

if SEX_cate=1 then SEX_cate="Male";
if SEX_cate=2 then SEX_cate="Female";


if RACENEW = 100 then RACENEW_cate = "White only";
   else if RACENEW = 200 then RACENEW_cate = "Black/African American
only";
   else if RACENEW = 300 then RACENEW_cate = "American Indian/Alaska
Native only";
   else if RACENEW = 400 then RACENEW_cate = "Asian only";
   else if RACENEW = 500 then RACENEW_cate = "Other Race and Multiple
Race";
   else if RACENEW = 510 then RACENEW_cate = "Other Race and Multiple
Race (2019-forward: Excluding American Indian/Alaska Native)";
   else if RACENEW = 520 then RACENEW_cate = "Other Race";
   else if RACENEW = 530 then RACENEW_cate = "Race Group Not
Releasable";
   else if RACENEW = 540 then RACENEW_cate = "Multiple Race";
   else if RACENEW = 541 then RACENEW_cate = "Multiple Race (1999-
2018: Including American Indian/Alaska Native)";
   else if RACENEW = 542 then RACENEW_cate = "American Indian/Alaska
Native and Any Other Race";

/*Income*/

 if INCFAM97ON2 = 10 then INCFAM97ON2_cate = "$0 - $34,999";
   else if INCFAM97ON2 = 20 then INCFAM97ON2_cate = "$35,000 -
$74,999";
   else if INCFAM97ON2 = 30 then INCFAM97ON2_cate = "$75,000 and
over";
```

```
   else if INCFAM97ON2 = 31 then INCFAM97ON2_cate = "$75,000 -
$99,999";
   else if INCFAM97ON2 = 32 then INCFAM97ON2_cate = "$100,000 and
over";
   else if INCFAM97ON2 = 96 then INCFAM97ON2_cate = "$20,000 or more
(no detail)";
   else if missing(INCFAM97ON2) then INCFAM97ON2_cate = "Missing";
   else INCFAM97ON2_cate = "Unknown";


/*Sleep quality*/
if SLEEPREST_f=00 then SLEEPREST_f="Never felt rested this past week";

/*Insurance*/

  if HINOTCOVE = 1 then HINOTCOVE_cate = "No, has coverage";
   else if HINOTCOVE = 2 then HINOTCOVE_cate = "Yes, has no coverage";
   else if missing(HINOTCOVE) then HINOTCOVE_cate = "Missing";
   else HINOTCOVE_cate = "Unknown";

run;



data NHIS;
      set NHIS;
      length SLEEPREST_cate $20.;

      if SLEEPREST = 0 or (1 <= SLEEPREST and SLEEPREST <= 2) then
            SLEEPREST_cate = "Low Quality";
      else if 3 <= SLEEPREST <= 5 then
            SLEEPREST_cate = "Medium Quality";
      else if 6 <= SLEEPREST <= 7 then
            SLEEPREST_cate = "High Quality";
      else SLEEPREST_cate = "Missing";
run;




/*Table 1 for association between primary predictor of interest and
confounder*/
data try;
set NHIS;
length ALCDAYSYR_cate $30.;
if ALCDAYSYR=0 then ALCDAYSYR_cate="Non-drinker";
else if 1<=ALCDAYSYR<=80 then ALCDAYSYR_cate="Sometime Drinker";
else if 81 <= ALCDAYSYR <= 150 then ALCDAYSYR_cate = "Moderate
drinker";
else if 151 <= ALCDAYSYR <= 312 then ALCDAYSYR_cate = "Frequent
drinker";
```

```
else if 313 <= ALCDAYSYR <= 365 then ALCDAYSYR_cate = "Daily drinker";
else ALCDAYSYR_cate = "Unknown";
run;


proc surveyfreq data=NHIS;;
 strata STRATA;
   cluster PSU;
   weight SAMPWEIGHT;
   table alcdaysyr;
run;


data try;
set try;
if SMOKEV=01 or SMOKEV=02;
run;

data try;
set try;
length ALCDAYSYR_cate $30.;
if ALCDAYSYR=0 then ALCDAYSYR_cate="Non-drinker";
else if 1<=ALCDAYSYR<=80 then ALCDAYSYR_cate="Sometime Drinker";
else if 81 <= ALCDAYSYR <= 150 then ALCDAYSYR_cate = "Moderate
drinker";
else if 151 <= ALCDAYSYR <= 312 then ALCDAYSYR_cate = "Frequent
drinker";
else if 313 <= ALCDAYSYR <= 365 then ALCDAYSYR_cate = "Daily drinker";
else ALCDAYSYR_cate = "Unknown";
run;


data try;
set try;
if SMOKFREQNOW_cate=1 then SMOKFREQNOW_cate="None-Smoker";
else if smokev=01 then SMOKFREQNOW_cate="None-Smoker";
else if SMOKFREQNOW_cate=2 then SMOKFREQNOW_cate="Some Day Smoker";
else if SMOKFREQNOW_cate=3 then SMOKFREQNOW_cate="Everyday Smoker";
run;

/*only run following data step once*/
data cate.NewReduceSet_160k;
set work.Reduced_due_to_missing;
run;


proc surveyfreq data=cate.NewReduceSet_160k;
 strata STRATA;
   cluster PSU;
   weight SAMPWEIGHT;
   table depfreq;
run;
```

```
/*Alcohol freq vs employment status*/

ods output CrossTabs=cate.alcohol_Emp;

proc surveyfreq data=cate.NewReduceSet_160k;
 strata STRATA;
   cluster PSU;
   weight SAMPWEIGHT;
   table ALCDAYSYR_cate*EMPSTATIMP5_cate/ row chisq;
run;
ods output close;



data see12;
set cate.alcohol_Emp;
keep ALCDAYSYR_cate EMPSTATIMP5_cate freq_perc cate_perc;
run;

proc surveylogistic data=cate.NewReduceSet_160k;
  strata STRATA;
  cluster PSU;
  weight SAMPWEIGHT;
  class ALCDAYSYR_cate(ref='Daily drinker')
EMPSTATIMP5_cate(ref='Employed') / param=ref;
  model ALCDAYSYR_cate = EMPSTATIMP5_cate;
run;



%OutTablePerc(cate.alcohol_Emp)

data alcohol_Emp;
set alcohol_Emp;
keep  ALCDAYSYR_cate F_EMPSTATIMP5_cate freq_perc cate_perc;
run;

/*Alcohol freq vs SEX*/
ods output CrossTabs=cate.alcohol_sex;

proc surveyfreq data=cate.NewReduceSet_160k;
 strata STRATA;
   cluster PSU;
   weight SAMPWEIGHT;
   table ALCDAYSYR_cate*SEX_cate/ row chisq;
run;

ods output close;

%OutTablePerc(cate.alcohol_sex)
```

```
data see6;
set cate.alcohol_sex;
keep ALCDAYSYR_cate SEX_cate freq_perc cate_perc;
run;



proc surveylogistic data=cate.NewReduceSet_160k;
  strata STRATA;
  cluster PSU;
  weight SAMPWEIGHT;
  class ALCDAYSYR_cate(ref='Daily drinker') SEX_cate(ref='Female') /
param=ref;
  model ALCDAYSYR_cate = SEX_cate;
run;


/*Alcohol freq vs Race*/
ods output CrossTabs=cate.alcohol_race;
proc surveyfreq data=cate.NewReduceSet_160k;
 strata STRATA;
   cluster PSU;
   weight SAMPWEIGHT;
   table ALCDAYSYR_cate*Racenew_cate/ row chisq;
run;
ods output close;

%OutTablePerc(cate.alcohol_race)

data seeRace;
set cate.alcohol_race;
keep ALCDAYSYR_cate RACENEW_cate freq_perc cate_perc;
run;

proc sort data=seeRace;
by ALCDAYSYR_cate freq_perc cate_perc ;
run;
/*Alcohol freq vs family income*/
ods output CrossTabs=cate.alcohol_income;

proc surveyfreq data=cate.NewReduceSet_160k;
 strata STRATA;
   cluster PSU;
   weight SAMPWEIGHT;
   table ALCDAYSYR_cate*incfam97on2_cate/ row chisq;
run;


ods output close;

%OutTablePerc(cate.alcohol_income)
```

```
data see4;
set cate.alcohol_income;
keep ALCDAYSYR_cate INCFAM97ON2_cate freq_perc cate_perc;
run;


proc surveyfreq data=see5;
strata STRATA;
   cluster PSU;
   weight SAMPWEIGHT;
table INCFAM97ON2_cate;
run;

/*Alcohol and insurance status*/

ods output CrossTabs=cate.Insur_sta;
proc surveyfreq data=cate.NewReduceSet_160k;
 strata STRATA;
   cluster PSU;
   weight SAMPWEIGHT;
   table ALCDAYSYR_cate*hinotcove_cate/ row chisq;
run;

ods output close;

data cate.insur_sta;
set cate.insur_sta;
freq_perc=catx('
',put(Frequency,10.),'('||strip(put(RowPercent,8.2))||'%)');
cate_perc=catx('
',put(Frequency,10.),'('||strip(put(Percent,8.2))||'%)');
run;

data see6;
set cate.insur_sta;
keep ALCDAYSYR_cate HINOTCOVE_cate freq_perc cate_perc;
run;



/*Alcohol Age*/
proc surveyfreq data=cate.NewReduceSet_160k;
      strata STRATA;
      cluster PSU;
      weight SAMPWEIGHT;
      table age;
run;

data cate.forAge_only;
set cate.NewReduceSet_160k;
length age_cate $20.;
```

```
 if 18 <= age <= 29 then age_cate = '18-29';
    else if 30 <= age <= 39 then age_cate = '30-39';
    else if 40 <= age <= 49 then age_cate = '40-49';
    else if 50 <= age <= 59 then age_cate = '50-59';
    else if 60 <= age <= 69 then age_cate = '60-69';
    else if 70 <= age <= 79 then age_cate = '70-79';
    else if 80 <= age <= 85 then age_cate = '80-85';
     else age_cate="Missing";
run;


ods output crosstabs=cate.alcohol_age;
proc surveyfreq data=cate.forage_only;
 strata STRATA;
   cluster PSU;
   weight SAMPWEIGHT;
   table ALCDAYSYR_cate*age_cate/ row chisq;
run;
ods output close;

%OutTablePerc(cate.alcohol_age)
data see8;
set cate.alcohol_age;
keep ALCDAYSYR_cate AGE_cate freq_perc cate_perc;
run;


/*Alcohol Sleep*/

/*ods output crosstabs=cate.alcohol_sleep;*/
/*proc surveyfreq data=cate.NewReduceSet_160k;*/
/* strata STRATA;*/
/*   cluster PSU;          */
/*   weight SAMPWEIGHT; */
/*   table ALCDAYSYR_cate*hrsleep_cate;*/
/*run;*/

data cate.NewReduceSet_160k;
set cate.NewReduceSet_160k;
if hrsleep=00 or hrsleep=25 or hrsleep=97
or hrsleep=98 or hrsleep=99 then hrsleep=.;

length hrsleep_cate $30.;
if hrsleep in (01,02,03,04,05) then hrsleep_cate="Short";
else if hrsleep in (06,07,08) then hrsleep_cate="Normal";
else hrsleep_cate="Long";
run;

proc surveyfreq data=cate.newreduceset_160k;
strata strata;
weight sampweight;
cluster psu;
```

```
table ALCDAYSYR_cate*hrsleep_cate;
run;

proc surveyfreq data=cate.nhis_00006;
strata strata;
weight sampweight;
cluster psu;
table age;
run;


/*alcohol sleeping*/

ods output crosstabs=cate.alcohol_sleep;
proc surveyfreq data=cate.NewReduceSet_160k;
 strata STRATA;
   cluster PSU;
   weight SAMPWEIGHT;
   table ALCDAYSYR_cate*hrsleep_cate/ row chisq;
/*  table SMOKFREQNOW_cate/ row chisq;*/

run;
ods output close;

%OutTablePerc(cate.alcohol_sleep)


data seeAlcohol_sleep;
set cate.alcohol_sleep;
keep ALCDAYSYR_cate hrsleep_cate freq_perc cate_perc;
run;
/*End of table 1 content*/



/*Break alcohol var into levels, use the categorical version instead*/
/*Still need to present it in numeric way result section*/

proc surveyreg data=NHIS;
   strata STRATA;
   cluster PSU;
   weight SAMPWEIGHT;
   class HINOTCOVE;
   model ALCDAYSYR = HINOTCOVE;
run;

proc surveyreg data=NHIS;
   strata STRATA;
   cluster PSU;
   weight SAMPWEIGHT;
   class HINOTCOVE;
   model ALCDAYSYR = HINOTCOVE;
```

```
   lsmeans HINOTCOVE / cl;
run;



proc surveylogistic data=NHIS nomcar;
title1 'Logistic looking at effect of sex on depression';
class sex(ref='Female')/param=reference;
model DEPFEELEVL=sex ALCDAYSYR sex*ALCDAYSYR;
strata STRATA;
cluster PSU;
weight SAMPWEIGHT;
run;



/*Table 2 start*/

data cate.Tb2;
set try;
run;

data cate.tb2;
set cate.tb2;
/*if DEPFREQ=0 or DEPFREQ=7 or DEPFREQ=8 or DEPFREQ=9*/
/*then depfreq=5;*/
run;

data Reduced_due_to_missing;
set cate.tb2;
if not missing (DEPFREQ);
run;

proc surveyfreq data=cate.nhis_00006;
  strata strata;
  cluster psu;
  weight sampweight;
  tables depfreq / chisq row;
run;


/*depression alcohol*/

ods output crosstabs=depression_alcohol;

proc surveyfreq data=Reduced_due_to_missing;
  strata strata;
  cluster psu;
  weight sampweight;
  tables depfreq*ALCDAYSYR_cate / chisq row;
run;
```

```
ods output close;


%OutTablePerc(depression_alcohol)

data depression_alcohol;
set depression_alcohol;
keep depfreq ALCDAYSYR_cate freq_perc cate_perc;
run;

/*depression employment*/
ods output crosstabs=depression_employment;
proc surveyfreq data=Reduced_due_to_missing;
  strata strata;
  cluster psu;
  weight sampweight;
  tables depfreq*EMPSTATIMP5_cate / chisq row;
run;
ods output close;

%OutTablePerc(depression_employment)

data depression_employment;
set depression_employment;
keep depfreq EMPSTATIMP5_cate freq_perc cate_perc;
run;



/*deprssion race*/
ods output crosstabs=depression_race;
proc surveyfreq data=Reduced_due_to_missing;
  strata strata;
  cluster psu;
  weight sampweight;
  tables depfreq*RACENEW_cate / chisq row;
run;
ods output close;

%OutTablePerc(depression_race)

data depression_race;
set depression_race;
keep depfreq race_cate freq_perc cate_perc;
run;



/*Depression income*/
ods output crosstabs=depression_income;

proc surveyfreq data=cate.newreduceset_160k;
```

```
  strata strata;
  cluster psu;
  weight sampweight;
  tables depfreq*INCFAM97ON2_cate/ chisq row;
run;

ods output close;


%OutTablePerc(depression_income)

data depression_income;
set depression_income;
keep depfreq INCFAM97ON2_cate  freq_perc cate_perc;
run;

/*depression age*/

data Reduced_due_to_missing;
set Reduced_due_to_missing;
 if 18 <= age <= 29 then age_cate = '18-29';
    else if 30 <= age <= 39 then age_cate = '30-39';
    else if 40 <= age <= 49 then age_cate = '40-49';
    else if 50 <= age <= 59 then age_cate = '50-59';
    else if 60 <= age <= 69 then age_cate = '60-69';
    else if 70 <= age <= 79 then age_cate = '70-79';
    else if 80 <= age <f= 85 then age_cate = '80-85';
     else age_cate="Missing";
run;

ods output crosstabs=depression_age;

proc surveyfreq data=Reduced_due_to_missing;
  strata strata;
  cluster psu;
  weight sampweight;
  tables depfreq*AGE_cate/chisq row;
run;
ods output close;

%OutTablePerc(depression_age)

data depression_age;
set depression_age;
keep depfreq age_cate freq_perc cate_perc;
run;


/*depression sex*/

ods output crosstabs=depression_sex;
```

```
proc surveyfreq data=Reduced_due_to_missing;
  strata strata;
  cluster psu;
  weight sampweight;
  tables depfreq*sex_cate/chisq row;
run;

ods output close;
%OutTablePerc(depression_sex)

data depression_sex;
set depression_sex;
keep depfreq sex_cate freq_perc cate_perc;
run;

/*depression insurance*/
ods output crosstabs=depression_insurance;


proc surveyfreq data=Reduced_due_to_missing;
  strata strata;
  cluster psu;
  weight sampweight;
  tables depfreq*hinotcove_cate/chisq row;
/*table HINOTCOVE_cate;*/
run;

ods output close;

%OutTablePerc(depression_insurance)

data depression_insurance;
set depression_insurance;
keep depfreq HINOTCOVE_cate freq_perc cate_perc;
run;



ods output close;


proc surveyfreq data=forSleep_reduced;
 strata strata;
  cluster psu;
  weight sampweight;
  table hrsleep_cate;
  run;

data forSleep_reduced;
set reduced_due_to_missing;
length hrsleep_cate $30.;
if hrsleep in (01,02,03,04,05) then hrsleep_cate="Short";
```

```
else if hrsleep in (06,07,08) then hrsleep_cate="Normal";
else hrsleep_cate="Long";
run;


/*Depression sleep*/
ods output crosstabs=depression_sleep;

proc surveyfreq data=forSleep_reduced;
  strata strata;
  cluster psu;
  weight sampweight;
  tables depfreq*hrsleep_cate/chisq row;
run;
ods output close;

%OutTablePerc(depression_sleep)

data depression_sleep;
set depression_sleep;
keep depfreq hrsleep_cate freq_perc cate_perc;
run;


proc surveyfreq data=cate.newreduceset_160k;
strata strata;
  cluster psu;
  weight sampweight;
  table depfreq/row;
  run;

proc surveyfreq data=cate.tb2;
strata strata;
  cluster psu;
  weight sampweight;
  table depfreq/row;
  run;

%macro SurvFreq(set,var);
proc surveyfreq data=&set;
strata strata;
  cluster psu;
  weight sampweight;
  table &var/row;
  run;
%mend;


/*Models Fitting Start*/
/*Start of handling proportional/adjacent odd*/
/*Fit only with response and predictor, assumption */
```

```
/*Proportional odds*/

proc contents data=cate.NewReduceSet_160k;
run;

proc logistic data=cate.NewReduceSet_160k;
   class ALCDAYSYR_cate / param=ref;
   model DEPFREQ = ALCDAYSYR_cate / link=logit aggregate scale=none
unequalslopes;
run;

/*POM main effect see if hold*/
proc logistic data=cate.NewReduceSet_160k;
   class ALCDAYSYR_cate(ref='Non-drinker') / param=ref;
   model depfreq_new = ALCDAYSYR_cate / link=clogit clodds=wald;
run;
/*hold,AIC 368214.76 */


/*hold AIC does not improve much, stick to POM*/

data cate.newreduceset_160k;
set cate.newreduceset_160k;
length depfreq_new $30;
if depfreq=1 then depfreq_new="High";
else if DEPFREQ in (2,3) then depfreq_new="Medium";
else if depfreq in (4,5) then depfreq_new="Low";
run;

proc surveyfreq data=cate.finalSet;
strata strata;
  cluster psu;
  weight sampweight;
  table AGE_cate;run;


proc logistic data=cate.NewReduceSet_160k;
   class ALCDAYSYR_cate(ref='Non-drinker') / param=ref;
   model depfreq_new = ALCDAYSYR_cate / link=logit aggregate
scale=none;
run;

/*POM was found to be violated, no partial pom available in sas,
opt in binary logistic regression model*/


/*Try Binary*/

/*data cate.binaryDepre;*/
/*set cate.newreduceset_160k;*/
/*length depfreq_binary $30;*/
/*if depfreq=1 then depfreq_binary="Daily";*/
```

```
/*else if DEPFREQ in (2,3,4,5) then depfreq_binary="Non-Daily";*/
/*run;*/




proc surveylogistic data=cate.binaryDepre;
   strata strata;
   cluster psu;
   weight sampweight;

   class
      ALCDAYSYR_cate(ref='Non-drinker')
/*       AGEGROUP_cate*/
       hrsleep_cate(ref='Short')
      SEX_cate (ref='Male')
      RACENEW_cate (ref='White only')
      EMPSTATIMP5_cate (ref='Unemployed')
      HINOTCOVE_cate (ref='Yes, has no coverage')
      INCFAM97ON2_cate (ref='$0 - $34,999')
      / param=ref;

   model depfreq_binary =
      ALCDAYSYR_cate
/*       AGEGROUP_cate*/
      SEX_cate
       hrsleep_cate
      RACENEW_cate
      EMPSTATIMP5_cate
      HINOTCOVE_cate
      INCFAM97ON2_cate
      / link=logit clparm;

run;

proc surveyfreq data=cate.newreduceset_160k;
strata strata;
   cluster psu;
   weight sampweight;
table depfreq;
run;



%SurvFreq(cate.NewReduceSet_160k,ALCDAYSYR_cate)



data cate.finalSet;
set cate.newreduceset_160k;
 if 18 <= age <= 29 then age_cate = '18-29';
    else if 30 <= age <= 39 then age_cate = '30-39';
```

```
    else if 40 <= age <= 49 then age_cate = '40-49';
    else if 50 <= age <= 59 then age_cate = '50-59';
    else if 60 <= age <= 69 then age_cate = '60-69';
    else if 70 <= age <= 79 then age_cate = '70-79';
    else if 80 <= age <= 85 then age_cate = '80-85';
     else age_cate="Missing";
run;


data cate.finalset;
set cate.finalset;

 /* Create individual flags */
    flag_age  = (Age_cate = "Missing");
    flag_sex  = (Sex_cate = "Missing");
    flag_race = (racenew_cate = "Missing");
    flag_emp  = (EMPSTATIMP5_cate = "Missing");
    flag_ins  = (HINOTCOVE_cate = "Missing");
    flag_sleep = (hrsleep_cate = "Missing");
    flag_income = (INCFAM97ON2_cate = "Missing");

    /* Create master flag: 1 = any missing, 0 = all complete */
    flag = max(of flag_:);
/*Individual flag for each var*/



run;


data cate.finalset;
set cate.finalset;
length depfreq_binary $30;
if depfreq in (1,2,3) then depfreq_binary="Severe";
else if depfreq in (4,5) then depfreq_binary="Not Severe";
run;

proc surveyfreq data=cate.finalset;
/*    where flag_income = 0;*/
    tables depfreq_binary;
run;


/*Including missing employment */
proc surveylogistic data=cate.finalSet;

strata strata;
   cluster psu;
   weight sampweight;
/*   where flag_income=0;*/
   class
      ALCDAYSYR_cate(ref='Non-drinker')
```

```
      age_cate
      SEX_cate
      RACENEW_cate
      EMPSTATIMP5_cate
      HINOTCOVE_cate
       hrsleep_cate
      INCFAM97ON2_cate / param=ref;

   model depfreq_binary =
      ALCDAYSYR_cate
      age_cate
      SEX_cate
      RACENEW_cate
      EMPSTATIMP5_cate
      HINOTCOVE_cate
      INCFAM97ON2_cate
       hrsleep_cate
      /link=logit clparm;
run;

/*Excluding missing employment*/
proc surveylogistic data=cate.finalSet;
     where flag_income=0;

     strata strata;
   cluster psu;
   weight sampweight;

   class
      ALCDAYSYR_cate(ref='Non-drinker')
      age_cate
      SEX_cate
      RACENEW_cate
      EMPSTATIMP5_cate
      HINOTCOVE_cate
       hrsleep_cate
      INCFAM97ON2_cate / param=ref;

   model depfreq_binary =
      ALCDAYSYR_cate
      age_cate
      SEX_cate
      RACENEW_cate
      EMPSTATIMP5_cate
      HINOTCOVE_cate
      INCFAM97ON2_cate
       hrsleep_cate
      / clparm;
run;


proc surveyfreq data=cate.finalset;
```

```
/*where flag_emp=0;*/

      strata strata;
   cluster psu;
   weight sampweight;
   table hrsleep_cate;
   run;




/*Including missing employment*/
   proc surveylogistic data=cate.finalSet;
/*    where flag_emp=0;*/

      strata strata;
   cluster psu;
   weight sampweight;

   class
      ALCDAYSYR_cate(ref='Non-drinker')
      age_cate
      SEX_cate
      RACENEW_cate
      EMPSTATIMP5_cate
      HINOTCOVE_cate
       hrsleep_cate
      INCFAM97ON2_cate / param=ref;

   model depfreq_binary =
      ALCDAYSYR_cate
      age_cate
      SEX_cate
      RACENEW_cate
      EMPSTATIMP5_cate
      HINOTCOVE_cate
      INCFAM97ON2_cate
       hrsleep_cate
      / clparm;
run;


/*excluding missing employment*/
   proc surveylogistic data=cate.finalSet;
      where flag_emp=0;

      strata strata;
   cluster psu;
   weight sampweight;

   class
      ALCDAYSYR_cate(ref='Non-drinker')
```

```
      age_cate
      SEX_cate
      RACENEW_cate
      EMPSTATIMP5_cate
      HINOTCOVE_cate
       hrsleep_cate
      INCFAM97ON2_cate / param=ref;

   model depfreq_binary =
      ALCDAYSYR_cate
      age_cate
      SEX_cate
      RACENEW_cate
      EMPSTATIMP5_cate
      HINOTCOVE_cate
      INCFAM97ON2_cate
       hrsleep_cate
      / clparm;
run;


data cate.finalset;
set cate.finalset;
Flag_emp_income=1;
if EMPSTATIMP5_cate="Missing" or INCFAM97ON2_cate="Missing"
then Flag_emp_income=0;
run;

/*the output domain flag=1 means?*/
/*the output domain flag=0 means?*/

/*Final Final Binary and unweighed partial proportional logistic*/
/*Model 1 and 2 without var but not keep/keep missing*/

/*Model 1,excluding income and employment, full data*/
proc surveylogistic data=cate.finalset;
strata strata;
   cluster psu;
   weight sampweight;

   class
      ALCDAYSYR_cate(ref='Non-drinker')
      age_cate(ref='18-29')
      SEX_cate(ref='Male')
      RACENEW_cate(ref='White only')
      EMPSTATIMP5_cate(ref='Unemployed')
      HINOTCOVE_cate(ref='Yes, has no coverage')
      hrsleep_cate(ref='Short')
      INCFAM97ON2_cate(ref='$75,000 - $99,999') / param=ref;

   model depfreq_binary(event='Severe')=
      ALCDAYSYR_cate
```

```
        age_cate
        SEX_cate
        RACENEW_cate
/*        EMPSTATIMP5_cate*/
        HINOTCOVE_cate
/*        INCFAM97ON2_cate*/
        hrsleep_cate
        / clodds;
run;
```

Number of Observations Read	164532
Number of Observations Used	164532
Sum of Weights Read	1.2804E9
Sum of Weights Used	1.2804E9

Response Profile			
Ordered Value	depfreq_binary	Total Frequency	Total Weight
1	Not Severe	138470	1087861585.1
2	Severe	26062	192572400.15

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	1.08426E9	1.04633E9
SC	1.08426E9	1.04633E9
-2 Log L	1.08426E9	1.04633E9

Testing Global Null Hypothesis: BETA=0				
Test	F Value	Num DF	Den DF	Pr > F
Likelihood Ratio	192.38	23.1538	33874	<.0001
Score	90.29	24	1440	<.0001
Wald	118.04	24	1440	<.0001
NOTE: Second-order Rao-Scott design correction 0.0365 applied to the Likelihood Ratio test.				

Type 3 Analysis of Effects				
Effect	F Value	Num DF	Den DF	Pr > F
ALCDAYSYR_cate	73.96	5	1459	<.0001
AGE_cate	36.47	7	1457	<.0001
SEX_cate	280.77	2	1462	<.0001
RACENEW_cate	45.70	6	1458	<.0001
HINOTCOVE_cate	36.97	2	1462	<.0001
hrsleep_cate	784.21	2	1462	<.0001

Analysis of Maximum Likelihood Estimates					
Parameter		Estimate	Standard Error	t Value	Pr > |t|
Intercept		-0.3967	0.0474	-8.37	<.0001
ALCDAYSYR_cate	Daily drinker	-0.1781	0.0492	-3.62	0.0003
ALCDAYSYR_cate	Frequent drinker	-0.2379	0.0444	-5.36	<.0001
ALCDAYSYR_cate	Moderate drinker	-0.5040	0.0462	-10.91	<.0001
ALCDAYSYR_cate	Sometime Drinker	-0.3658	0.0283	-12.91	<.0001
ALCDAYSYR_cate	Unknown	-0.5799	0.0323	-17.95	<.0001
AGE_cate	30-39	-0.2104	0.0301	-6.99	<.0001
AGE_cate	40-49	-0.1127	0.0324	-3.48	0.0005
AGE_cate	50-59	-0.0884	0.0309	-2.86	0.0043
AGE_cate	60-69	-0.2708	0.0315	-8.59	<.0001
AGE_cate	70-79	-0.4771	0.0392	-12.17	<.0001
AGE_cate	80-85	-0.4893	0.0465	-10.52	<.0001
AGE_cate	Missing	-0.4040	0.4650	-0.87	0.3851
SEX_cate	Female	0.4537	0.0191	23.70	<.0001
SEX_cate	Missing	0.9486	1.1858	0.80	0.4239
RACENEW_cate	American Indian/Alaska Native	-0.0232	0.0967	-0.24	0.8107
RACENEW_cate	Asian only	-0.6252	0.0498	-12.56	<.0001
RACENEW_cate	Black/African American only	-0.2674	0.0296	-9.03	<.0001
RACENEW_cate	Missing	-0.4187	0.0833	-5.03	<.0001
RACENEW_cate	Multiple Race (1999-2018: Incl	0.2809	0.0721	3.90	0.0001
RACENEW_cate	Other Race	0.1395	0.1687	0.83	0.4085
HINOTCOVE_cate	Missing	-0.1818	0.1640	-1.11	0.2679
HINOTCOVE_cate	No, has coverage	-0.2360	0.0275	-8.59	<.0001
hrsleep_cate	Long	-0.4227	0.0348	-12.15	<.0001
hrsleep_cate	Normal	-1.0400	0.0285	-36.49	<.0001
NOTE: The degrees of freedom for the t tests is 1463.					

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	62.1	Somers' D	0.258
Percent Discordant	36.3	Gamma	0.262
Percent Tied	1.5	Tau-a	0.069
Pairs	3608805140	c	0.629

Odds Ratio Estimates and t Confidence Intervals				
Effect	Unit	Estimate	95% Confidence Limits	
ALCDAYSYR_cate Daily drinker vs Non-drinker	1.0000	0.837	0.760	0.922
ALCDAYSYR_cate Frequent drinker vs Non-drinker	1.0000	0.788	0.723	0.860
ALCDAYSYR_cate Moderate drinker vs Non-drinker	1.0000	0.604	0.552	0.661
ALCDAYSYR_cate Sometime Drinker vs Non-drinker	1.0000	0.694	0.656	0.733
ALCDAYSYR_cate Unknown vs Non-drinker	1.0000	0.560	0.526	0.597
AGE_cate 30-39 vs 18-29	1.0000	0.810	0.764	0.860
AGE_cate 40-49 vs 18-29	1.0000	0.893	0.838	0.952
AGE_cate 50-59 vs 18-29	1.0000	0.915	0.862	0.973
AGE_cate 60-69 vs 18-29	1.0000	0.763	0.717	0.811
AGE_cate 70-79 vs 18-29	1.0000	0.621	0.575	0.670
AGE_cate 80-85 vs 18-29	1.0000	0.613	0.560	0.672
AGE_cate Missing vs 18-29	1.0000	0.668	0.268	1.662
SEX_cate Female vs Male	1.0000	1.574	1.516	1.634
SEX_cate Missing vs Male	1.0000	2.582	0.252	26.436
RACENEW_cate American Indian/Alaska Native vs White only	1.0000	0.977	0.808	1.181
RACENEW_cate Asian only vs White only	1.0000	0.535	0.485	0.590
RACENEW_cate Black/African American only vs White only	1.0000	0.765	0.722	0.811
RACENEW_cate Missing vs White only	1.0000	0.658	0.559	0.775
RACENEW_cate Multiple Race (1999-2018: Incl vs White only	1.0000	1.324	1.150	1.525
RACENEW_cate Other Race vs White only	1.0000	1.150	0.826	1.601
HINOTCOVE_cate Missing vs Yes, has no coverage	1.0000	0.834	0.604	1.150
HINOTCOVE_cate No, has coverage vs Yes, has no coverage	1.0000	0.790	0.748	0.833
hrsleep_cate Long vs Short	1.0000	0.655	0.612	0.702
hrsleep_cate Normal vs Short	1.0000	0.353	0.334	0.374
NOTE: The degrees of freedom in computing the confidence limits is 1463.				

```
ods output crosstabs=model2;

/*Model 2 , domain analysis,only read domain flag=1  */
proc surveylogistic data=cate.finalset;
strata strata;
   cluster psu;
   weight sampweight;

   class
      ALCDAYSYR_cate(ref='Non-drinker')
      age_cate(ref='18-29')
      SEX_cate(ref='Male')
      RACENEW_cate(ref='White only')
      EMPSTATIMP5_cate(ref='Unemployed')
      HINOTCOVE_cate(ref='Yes, has no coverage')
      hrsleep_cate(ref='Short')
      INCFAM97ON2_cate(ref='$75,000 - $99,999') / param=ref;

   model depfreq_binary(event='Severe')=
      ALCDAYSYR_cate
```

```
        age_cate
        SEX_cate
        RACENEW_cate
        EMPSTATIMP5_cate
        HINOTCOVE_cate
        INCFAM97ON2_cate
        hrsleep_cate
        / clodds;
         domain Flag_emp_income;
run;
ods output close;
```

Domain Summary	
Number of Observations	164532
Number of Observations in Domain	124335
Number of Observations not in Domain	40197
Sum of Weights in Domain	958460493

Number of Observations Read	164532
Number of Observations Used	164532
Sum of Weights Read	9.5846E8
Sum of Weights Used	9.5846E8

Response Profile

Ordered Value	depfreq_binary	Total Frequency	Total Weight
1	Not Severe	104394	814170448
2	Severe	19941	144290045

Probability modeled is depfreq_binary='Severe'.

Model Fit Statistics

Criterion	Intercept Only	Intercept and Covariates
AIC	812105058	754553233
SC	812105077	754553813
-2 Log L	812105056	754553171

Testing Global Null Hypothesis: BETA=0

Test	F Value	Num DF	Den DF	Pr > F
Likelihood Ratio	235.67	28.5173	41721	<.0001
Score	90.26	30	1434	<.0001
Wald	132.38	30	1434	<.0001

NOTE: Second-order Rao-Scott design correction 0.0520 applied to the Likelihood Ratio test.

Type 3 Analysis of Effects

Effect	F Value	Num DF	Den DF	Pr > F
ALCDAYSYR_cate	62.62	5	1459	<.0001
AGE_cate	102.36	7	1457	<.0001
SEX_cate	133.27	2	1462	<.0001
RACENEW_cate	44.95	6	1458	<.0001
EMPSTATIMP5_cate	296.99	2	1462	<.0001
HINOTCOVE_cate	0.21	2	1462	0.8077
INCFAM97ON2_cate	250.32	4	1460	<.0001
hrsleep_cate	524.07	2	1462	<.0001

Analysis of Maximum Likelihood Estimates					
Parameter		Estimate	Standard Error	t Value	Pr > |t|
Intercept		-0.6706	0.0698	-9.60	<.0001
ALCDAYSYR_cate	Daily drinker	0.0566	0.0501	1.13	0.2589
ALCDAYSYR_cate	Frequent drinker	0.0641	0.0463	1.39	0.1658
ALCDAYSYR_cate	Moderate drinker	-0.2281	0.0482	-4.73	<.0001
ALCDAYSYR_cate	Sometime Drinker	-0.1639	0.0293	-5.59	<.0001
ALCDAYSYR_cate	Unknown	-0.5650	0.0368	-15.34	<.0001
AGE_cate	30-39	-0.0471	0.0361	-1.31	0.1919
AGE_cate	40-49	0.1296	0.0389	3.33	0.0009
AGE_cate	50-59	0.0723	0.0370	1.96	0.0505
AGE_cate	60-69	-0.3736	0.0397	-9.40	<.0001
AGE_cate	70-79	-0.8399	0.0500	-16.80	<.0001
AGE_cate	80-85	-1.0411	0.0598	-17.40	<.0001
AGE_cate	Missing	-0.8286	0.4091	-2.03	0.0430
SEX_cate	Female	0.3718	0.0228	16.31	<.0001
SEX_cate	Missing	1.2803	1.1885	1.08	0.2815
RACENEW_cate	American Indian/Alaska Native	-0.4346	0.1254	-3.46	0.0005
RACENEW_cate	Asian only	-0.5671	0.0580	-9.78	<.0001
RACENEW_cate	Black/African American only	-0.4302	0.0329	-13.06	<.0001
RACENEW_cate	Missing	-0.0517	0.2045	-0.25	0.8005
RACENEW_cate	Multiple Race (1999-2018: Incl	0.1966	0.0737	2.67	0.0077
RACENEW_cate	Other Race	0.1014	0.1425	0.71	0.4770
EMPSTATIMP5_cate	Employed	-0.6287	0.0266	-23.67	<.0001
EMPSTATIMP5_cate	Missing	0.1677	0.0582	2.88	0.0040
HINOTCOVE_cate	Missing	-0.0566	0.2292	-0.25	0.8051
HINOTCOVE_cate	No, has coverage	0.0184	0.0332	0.55	0.5794
INCFAM97ON2_cate	$0 - $34,999	0.6881	0.0396	17.36	<.0001
INCFAM97ON2_cate	$100,000 and over	-0.3075	0.0446	-6.89	<.0001
INCFAM97ON2_cate	$35,000 - $74,999	0.2398	0.0393	6.10	<.0001
INCFAM97ON2_cate	Missing	-0.1197	0.0507	-2.36	0.0183
hrsleep_cate	Long	-0.3837	0.0416	-9.23	<.0001
hrsleep_cate	Normal	-0.9046	0.0294	-30.80	<.0001
NOTE: The degrees of freedom for the t tests is 1463.					

Odds Ratio Estimates and t Confidence Intervals				
Effect	Unit	Estimate	95% Confidence Limits	
ALCDAYSYR_cate Daily drinker vs Non-drinker	1.0000	1.058	0.959	1.168
ALCDAYSYR_cate Frequent drinker vs Non-drinker	1.0000	1.066	0.974	1.168
ALCDAYSYR_cate Moderate drinker vs Non-drinker	1.0000	0.796	0.724	0.875
ALCDAYSYR_cate Sometime Drinker vs Non-drinker	1.0000	0.849	0.801	0.899
ALCDAYSYR_cate Unknown vs Non-drinker	1.0000	0.568	0.529	0.611
AGE_cate 30-39 vs 18-29	1.0000	0.954	0.889	1.024
AGE_cate 40-49 vs 18-29	1.0000	1.138	1.055	1.229
AGE_cate 50-59 vs 18-29	1.0000	1.075	1.000	1.156
AGE_cate 60-69 vs 18-29	1.0000	0.688	0.637	0.744
AGE_cate 70-79 vs 18-29	1.0000	0.432	0.391	0.476
AGE_cate 80-85 vs 18-29	1.0000	0.353	0.314	0.397
AGE_cate Missing vs 18-29	1.0000	0.437	0.196	0.974
SEX_cate Female vs Male	1.0000	1.450	1.387	1.517
SEX_cate Missing vs Male	1.0000	3.598	0.350	37.022
RACENEW_cate American Indian/Alaska Native vs White only	1.0000	0.648	0.506	0.828
RACENEW_cate Asian only vs White only	1.0000	0.567	0.506	0.636
RACENEW_cate Black/African American only vs White only	1.0000	0.650	0.610	0.694
RACENEW_cate Missing vs White only	1.0000	0.950	0.636	1.418
RACENEW_cate Multiple Race (1999-2018: Incl vs White only	1.0000	1.217	1.053	1.406
RACENEW_cate Other Race vs White only	1.0000	1.107	0.837	1.464
EMPSTATIMP5_cate Employed vs Unemployed	1.0000	0.533	0.506	0.562
EMPSTATIMP5_cate Missing vs Unemployed	1.0000	1.183	1.055	1.326
HINOTCOVE_cate Missing vs Yes, has no coverage	1.0000	0.945	0.603	1.482
HINOTCOVE_cate No, has coverage vs Yes, has no coverage	1.0000	1.019	0.954	1.087
INCFAM97ON2_cate $0 - $34,999 vs $75,000 - $99,999	1.0000	1.990	1.841	2.151
INCFAM97ON2_cate $100,000 and over vs $75,000 - $99,999	1.0000	0.735	0.674	0.803
INCFAM97ON2_cate $35,000 - $74,999 vs $75,000 - $99,999	1.0000	1.271	1.177	1.373
INCFAM97ON2_cate Missing vs $75,000 - $99,999	1.0000	0.887	0.803	0.980
hrsleep_cate Long vs Short	1.0000	0.681	0.628	0.739
hrsleep_cate Normal vs Short	1.0000	0.405	0.382	0.429
NOTE: The degrees of freedom in computing the confidence limits is 1463.				

```
/*Model 3, use full data set(include rows with missing
in employment and income),include emp and income var*/
proc surveylogistic data=cate.finalset;
strata strata;
   cluster psu;
   weight sampweight;

   class
      ALCDAYSYR_cate(ref='Non-drinker')
      age_cate(ref='18-29')
      SEX_cate(ref='Male')
      RACENEW_cate(ref='White only')
      EMPSTATIMP5_cate(ref='Unemployed')
      HINOTCOVE_cate(ref='Yes, has no coverage')
      hrsleep_cate(ref='Short')
      INCFAM97ON2_cate(ref='$75,000 - $99,999') / param=ref;

   model depfreq_binary(event='Severe')=
```

```
        ALCDAYSYR_cate
        age_cate
        SEX_cate
        RACENEW_cate
        EMPSTATIMP5_cate
        HINOTCOVE_cate
        INCFAM97ON2_cate
        hrsleep_cate
        / clodds;
run;

proc surveyfreq data=NHIS;
strata strata;
    cluster psu;
    weight sampweight;
    table depfreq;
    run;
```

Number of Observations Read	164532
Number of Observations Used	164532
Sum of Weights Read	1.2804E9
Sum of Weights Used	1.2804E9

Response Profile			
Ordered Value	depfreq_binary	Total Frequency	Total Weight
1	Not Severe	138470	1087861585.1
2	Severe	26062	192572400.15

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	1.08426E9	1.02023E9
SC	1.08426E9	1.02023E9
-2 Log L	1.08426E9	1.02023E9

Testing Global Null Hypothesis: BETA=0				
Test	F Value	Num DF	Den DF	Pr > F
Likelihood Ratio	260.44	28.7914	42122	<.0001
Score	103.27	30	1434	<.0001
Wald	136.05	30	1434	<.0001

NOTE: Second-order Rao-Scott design correction 0.0420 applied to the Likelihood Ratio test.

Type 3 Analysis of Effects				
Effect	F Value	Num DF	Den DF	Pr > F
ALCDAYSYR_cate	62.34	5	1459	<.0001
AGE_cate	95.47	7	1457	<.0001
SEX_cate	218.07	2	1462	<.0001
RACENEW_cate	54.79	6	1458	<.0001
EMPSTATIMP5_cate	286.53	2	1462	<.0001
HINOTCOVE_cate	1.07	2	1462	0.3423
INCFAM97ON2_cate	209.16	4	1460	<.0001
hrsleep_cate	566.78	2	1462	<.0001

Analysis of Maximum Likelihood Estimates					
Parameter		Estimate	Standard Error	t Value	Pr > |t|
Intercept		-0.5828	0.0650	-8.97	<.0001
ALCDAYSYR_cate	Daily drinker	0.0419	0.0492	0.85	0.3951
ALCDAYSYR_cate	Frequent drinker	0.0619	0.0455	1.36	0.1740
ALCDAYSYR_cate	Moderate drinker	-0.2221	0.0470	-4.73	<.0001
ALCDAYSYR_cate	Sometime Drinker	-0.1683	0.0287	-5.86	<.0001
ALCDAYSYR_cate	Unknown	-0.5591	0.0363	-15.39	<.0001
AGE_cate	30-39	-0.1226	0.0308	-3.98	<.0001
AGE_cate	40-49	0.000054	0.0333	0.00	0.9987
AGE_cate	50-59	-0.0175	0.0317	-0.55	0.5812
AGE_cate	60-69	-0.3813	0.0329	-11.58	<.0001
AGE_cate	70-79	-0.7719	0.0419	-18.41	<.0001
AGE_cate	80-85	-0.8629	0.0500	-17.25	<.0001
AGE_cate	Missing	-0.3876	0.4619	-0.84	0.4015
SEX_cate	Female	0.4046	0.0194	20.89	<.0001
SEX_cate	Missing	0.8271	1.1850	0.70	0.4853
RACENEW_cate	American Indian/Alaska Native	-0.1116	0.0994	-1.12	0.2619
RACENEW_cate	Asian only	-0.6074	0.0506	-12.01	<.0001
RACENEW_cate	Black/African American only	-0.3760	0.0294	-12.79	<.0001
RACENEW_cate	Missing	-0.4133	0.0837	-4.94	<.0001
RACENEW_cate	Multiple Race (1999-2018: Incl	0.2151	0.0723	2.97	0.0030
RACENEW_cate	Other Race	0.1190	0.1714	0.69	0.4875
EMPSTATIMP5_cate	Employed	-0.5899	0.0251	-23.47	<.0001
EMPSTATIMP5_cate	Missing	0.1464	0.0666	2.20	0.0282
HINOTCOVE_cate	Missing	0.0306	0.1674	0.18	0.8549
HINOTCOVE_cate	No, has coverage	-0.0401	0.0288	-1.39	0.1644
INCFAM97ON2_cate	$0 - $34,999	0.6538	0.0393	16.65	<.0001
INCFAM97ON2_cate	$100,000 and over	-0.2922	0.0445	-6.57	<.0001
INCFAM97ON2_cate	$35,000 - $74,999	0.2236	0.0393	5.69	<.0001
INCFAM97ON2_cate	Missing	-0.1252	0.0583	-2.15	0.0319
hrsleep_cate	Long	-0.4240	0.0404	-10.50	<.0001
hrsleep_cate	Normal	-0.9389	0.0291	-32.26	<.0001
NOTE: The degrees of freedom for the t tests is 1463.					

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	66.5	Somers' D	0.338
Percent Discordant	32.6	Gamma	0.341
Percent Tied	0.9	Tau-a	0.090
Pairs	3608805140	c	0.669

Odds Ratio Estimates and t Confidence Intervals				
Effect	Unit	Estimate	95% Confidence Limits	
ALCDAYSYR_cate Daily drinker vs Non-drinker	1.0000	1.043	0.947	1.148
ALCDAYSYR_cate Frequent drinker vs Non-drinker	1.0000	1.064	0.973	1.163
ALCDAYSYR_cate Moderate drinker vs Non-drinker	1.0000	0.801	0.730	0.878
ALCDAYSYR_cate Sometime Drinker vs Non-drinker	1.0000	0.845	0.799	0.894
ALCDAYSYR_cate Unknown vs Non-drinker	1.0000	0.572	0.532	0.614
AGE_cate 30-39 vs 18-29	1.0000	0.885	0.833	0.940
AGE_cate 40-49 vs 18-29	1.0000	1.000	0.937	1.067
AGE_cate 50-59 vs 18-29	1.0000	0.983	0.923	1.046
AGE_cate 60-69 vs 18-29	1.0000	0.683	0.640	0.729
AGE_cate 70-79 vs 18-29	1.0000	0.462	0.426	0.502
AGE_cate 80-85 vs 18-29	1.0000	0.422	0.383	0.465
AGE_cate Missing vs 18-29	1.0000	0.679	0.274	1.679
SEX_cate Female vs Male	1.0000	1.499	1.443	1.557
SEX_cate Missing vs Male	1.0000	2.287	0.224	23.374
RACENEW_cate American Indian/Alaska Native vs White only	1.0000	0.894	0.736	1.087
RACENEW_cate Asian only vs White only	1.0000	0.545	0.493	0.602
RACENEW_cate Black/African American only vs White only	1.0000	0.687	0.648	0.727
RACENEW_cate Missing vs White only	1.0000	0.661	0.561	0.780
RACENEW_cate Multiple Race (1999-2018: Incl vs White only	1.0000	1.240	1.076	1.429
RACENEW_cate Other Race vs White only	1.0000	1.126	0.805	1.577
EMPSTATIMP5_cate Employed vs Unemployed	1.0000	0.554	0.528	0.582
EMPSTATIMP5_cate Missing vs Unemployed	1.0000	1.158	1.016	1.319
HINOTCOVE_cate Missing vs Yes, has no coverage	1.0000	1.031	0.742	1.432
HINOTCOVE_cate No, has coverage vs Yes, has no coverage	1.0000	0.961	0.908	1.017
INCFAM97ON2_cate $0 - $34,999 vs $75,000 - $99,999	1.0000	1.923	1.780	2.077
INCFAM97ON2_cate $100,000 and over vs $75,000 - $99,999	1.0000	0.747	0.684	0.815
INCFAM97ON2_cate $35,000 - $74,999 vs $75,000 - $99,999	1.0000	1.251	1.158	1.351
INCFAM97ON2_cate Missing vs $75,000 - $99,999	1.0000	0.882	0.787	0.989
hrsleep_cate Long vs Short	1.0000	0.654	0.605	0.708
hrsleep_cate Normal vs Short	1.0000	0.391	0.369	0.414
NOTE: The degrees of freedom in computing the confidence limits is 1463.				